
Lasso Regression Solution

MSCS2202 – Challenge Assignment

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1. Overview

This report documents the end-to-end machine learning pipeline used to predict continuous target values from 15 numerical features (x0–x14) using Lasso Regression. The evaluation metric is R^2 .

Evaluation:

$R^2 = 1$	$R^2 = 0$	$R^2 < 0$
Perfect prediction	Predicting the mean	Worse than predicting the mean

2. Step 1 — Exploratory Data Analysis

Dataset	Rows	Columns	Notes
Training Set	2,500	16	x0–x14 features + target column
Test Set	2,500	15	x0–x14 features only

Target Variable Statistics

Statistic	Value	Notes
Mean	-22.04	Negative shift center
Std	1,978.18	Extremely high variance
Min	-41,008.02	Severe negative outlier
Max	69,628.20	Severe positive outlier

Key observation: the target has extreme outliers. These extreme rows dominate the squared error and distort model training.

3. Step 2 — Data Cleaning

3a. Handling Missing Values (Imputation)

Both the training and test files contain missing values in every feature column.

Method: Using the average value of the column to fill the empty space in the table.

3b. Outlier Removal

Problem: Linear regression minimizes the sum of squared errors. Squaring amplifies extreme errors.

The extreme rows dominate the entire training signal. The solution is to remove them from training only.

Threshold selection: 5 different thresholds were tested via cross-validation to find the optimal cutoff point.

Threshold	Rows Kept	Rows Removed	CV R ²
target > 2,000	2,444	56	0.0154
target > 3,000	2,461	39	0.0225
target > 4,000	2,469	31	0.0273
 target > 5,000	2,476	24	0.0316 ← best
target > 7,000	2,485	15	0.0286
target > 10,000	2,490	10	0.0143

Result: 24 rows removed. 2,476 rows used for training. All 2,500 test rows are still predicted.

4. Step 3 — Feature Scaling for Lasso Regression

StandardScaler: transforms each feature to mean=0 and std=1.

Reason: Lasso penalizes coefficients based on their magnitude. The larger scale feature may dominate the penalty and distort feature selection.

- fit_transform on training data (learns the mean and std of each feature)
- transform only on test data (applies the same training statistics)

5. Step 4 — Model Selection

5a. Ridge Regression vs. Lasso Regression

Ridge Regression and Lasso Regression models were tested using cross-validation with the same data cleaning pipeline (outlier threshold = 5,000):

Model	Best CV R ²	Notes
Ridge Regression	0.0310	Shrinks all coefficients, keeps all 15 features
Lasso Regression	0.0316	Shrinks weak features to zero -- auto feature selection

5b. Selected Lasso Regression

Lasso (Least Absolute Shrinkage and Selection Operator) uses an L1 penalty — it penalizes the sum of absolute coefficient values. This causes weak feature coefficients to shrink to exactly zero, effectively removing them from the model.

- Ridge (L2 penalty): shrinks all coefficients but never removes any feature
- Lasso (L1 penalty): sets weak coefficients to exactly zero; built-in feature selection

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \dots$$

On this dataset, many features have near-zero correlation with the target. Lasso automatically identified and removed 5 of them:

Feature	Coefficient	Status
x0	0.0000	Zeroed out
x1	0.0000	Zeroed out
x2	-6.2389	Kept
x3	-5.0738	Kept
x4	41.8812	Kept
x5	-19.2333	Kept
x6	5.4151	Kept
x7	0.0000	Zeroed out
x8	-11.0864	Kept
x9	90.9568	Kept — strongest signal
x10	0.4068	Kept
x11	0.0000	Zeroed out
x12	-0.5773	Kept
x13	0.0000	Zeroed out
x14	5.2090	Kept

Result: 10 of 15 features kept. 5 features (x0, x1, x7, x11, x13) zeroed out as irrelevant.

6. Step 5 — Hyperparameter Tuning (Alpha)

6a. Alpha (α)

Alpha is Lasso's regularization strength parameter -- a knob that controls how aggressively the model penalizes large coefficients.

$$Loss = \text{Sum of Squared Errors} + \alpha \sum_{j=1}^p |\beta_j|$$

Alpha	Effect	Risk
Too small	Fits training data closely	Overfitting
Just right	Learn real patterns	Best generalization
Too large	Model over-simplified	Underfitting

6b. Tuning Method — 5-Fold Cross-Validation

The training data is split into 5 equal folds. The model trains on 4 folds and validates on the remaining 1. This rotates 5 times so every row is validated exactly once. The alpha with the highest average R^2 is selected.

Alpha	CV R^2	Std Dev
0.1	0.0311	±0.0105
0.5	0.0311	±0.0106
1	0.0312	±0.0106
1.5	0.0313	±0.0107
2	0.0314	±0.0108
3	0.0315	±0.0109
5	0.0316 ← best	±0.0112
10	0.0304	±0.0111

Selected: $\alpha = 5$ with CV $R^2 = 0.0316$

7. Step 6 — Final Model & Predictions

- Final model trained on all 2,476 cleaned training rows using alpha = 5
- StandardScaler fitted on clean training data, then applied to test data
- Predictions generated for all 2,500 test rows

8. Conclusion

Lasso Regression with data cleaning achieved a CV $R^2 = 0.0316$, making it the best single-model approach tested.

8a. Why R^2 is Low

The low R^2 reflects a genuinely difficult dataset:

- Target standard deviation $\approx 1,978$ (driven by extreme outliers up to $\pm 69,000$)
- Raw features have very weak linear correlations with the target (max 0.23)
- A linear model has fundamental limits on non-linear patterns